Note 11.3 Polar Coordinates

Definition of Polar Coordinates

Definition

To define polar coordinates, we first fix an origin O (called the pole) and an initial ray from O. Then each point P can be located by assigning to it a polar coordinate pair (r, θ) in which r gives the directed distance from O to P and θ gives the directed angle from the initial ray to ray OP.

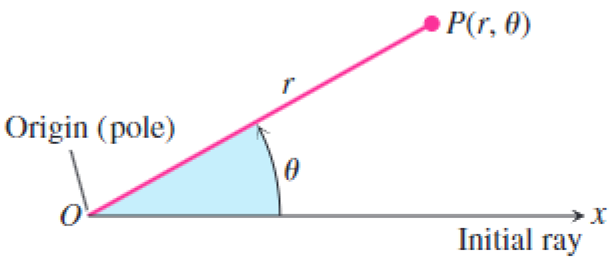

FIGURE 11.18 To define polar coordinates for the plane, we start with an origin, called the pole, and an initial ray.

Remarks

The polar coordinates are not unique. The θ depends on measuring by counterclockwise or clockwise(positive) and the r can be negative when measuring it by the going backward after deciding θ.

Example

For the polar coordinates of the point $P(2, \pi/6)$, it can be be represented as

$$\left(2, \frac{\pi}{6} + 2n\pi\right), n \in \mathbf{Z}$$

and

$$\left(-2, -\frac{5}{6}\pi + 2n\pi\right), n \in \mathbf{Z}$$

Polar Equations and Graphs

Equation	Graph
$r = a$	Circle of radius $\|a\|$ centered at O
$\theta = \theta_0$	Line through O making an angle θ_0 with the initial ray

Relating Polar and Cartesian Coordinates

Equations Relating Polar and Cartesian Coordinates

$$x = r\cos\theta, y = r\sin\theta, r^2 = x^2 + y^2, \tan\theta = \frac{y}{x}$$